

Vibration Energy Harvesting

Professor Yuji Suzuki

Department of Mechanical Engineering

The University of Tokyo, Japan

e-mail: ysuzuki@mesl.t.u-tokyo.ac.jp



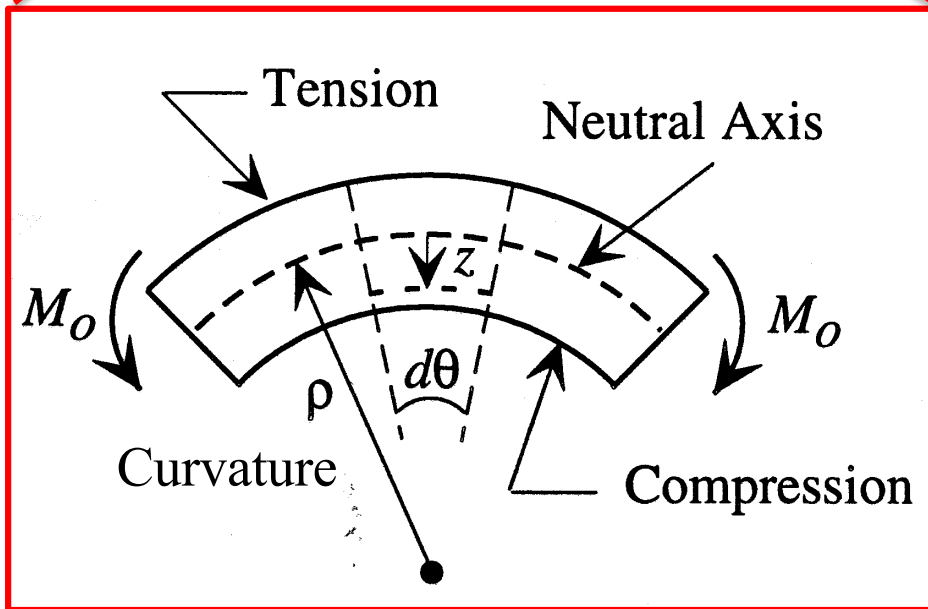
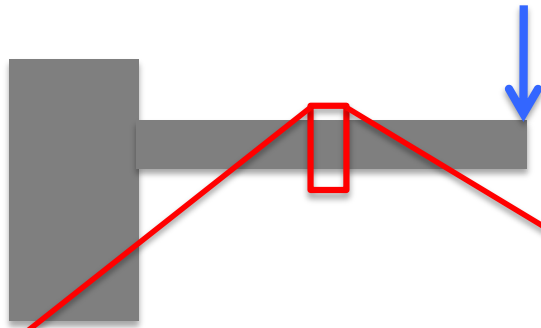
東京大学
THE UNIVERSITY OF TOKYO

Objective

Experience vibration energy harvesting using piezoelectric cantilever

- ✓ **Carry out a simple power generation experiment using the piezoelectric cantilever.**
- ✓ **Learn the resonant frequency and the impedance matching.**

Beam Theory



ρ : radius of curvature

$$dx = \rho d\theta$$

$$dL = (\rho - z) d\theta$$

$$= dx - \frac{z}{\rho} dx$$

$$\epsilon_x = -\frac{z}{\rho}$$

Strain

$$\sigma_x = -\frac{z}{\rho} E$$

Stress

$$M = \int_{-H/2}^{H/2} W z \sigma_x dz$$

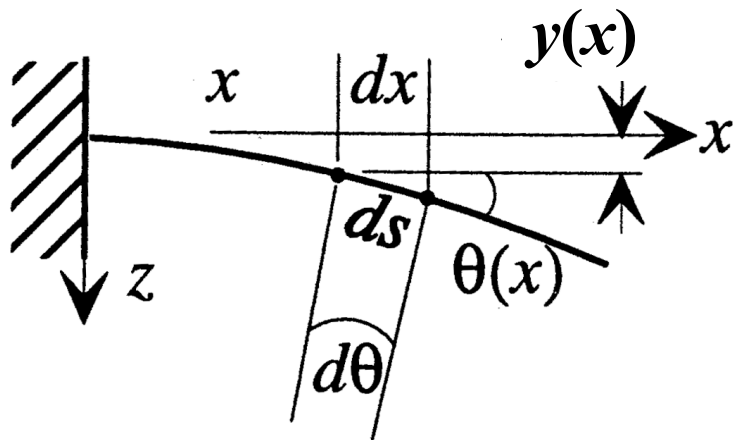
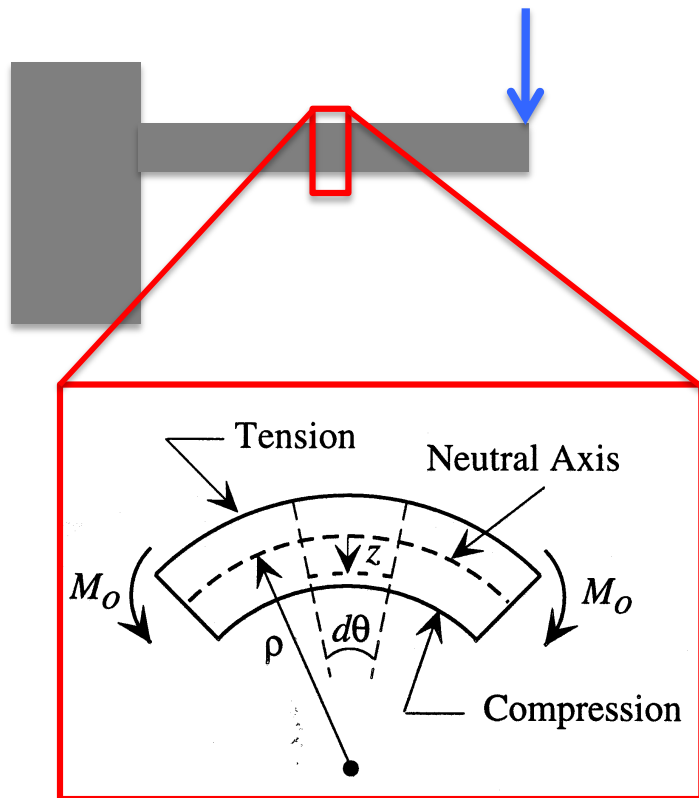
Moment

$$= -\frac{EI}{\rho}$$

$$I = \frac{1}{12} WH^3$$

Moment of
inertia

Linear Beam Theory



$$ds = \rho d\theta$$

$$ds = \frac{dx}{\cos \theta}$$

$$\frac{dy}{dx} = \tan \theta$$

$$\frac{d^2 y}{dx^2} = \frac{1}{\cos^2 \theta} \frac{d\theta}{dx}$$

$$\frac{1}{\rho} = \frac{d^2 y / dx^2}{\left\{ 1 + \left(\frac{dy}{dx} \right)^2 \right\}^{3/2}} \sim \frac{d^2 y}{dx^2}$$

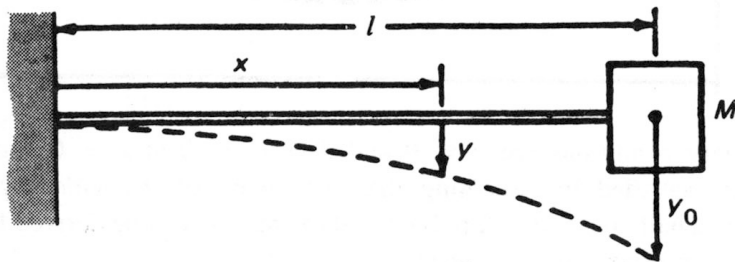
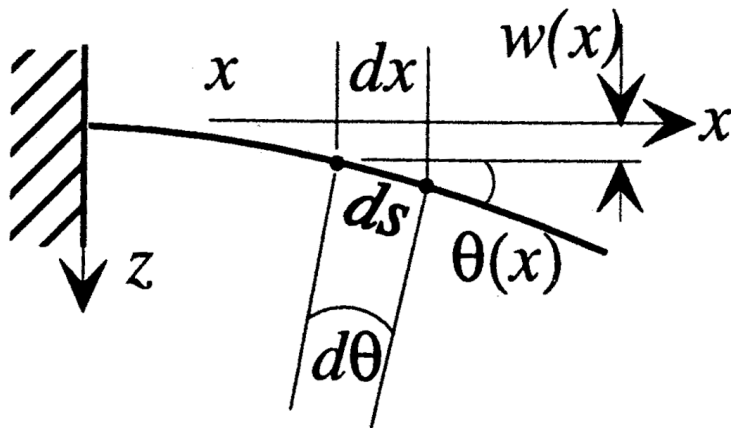
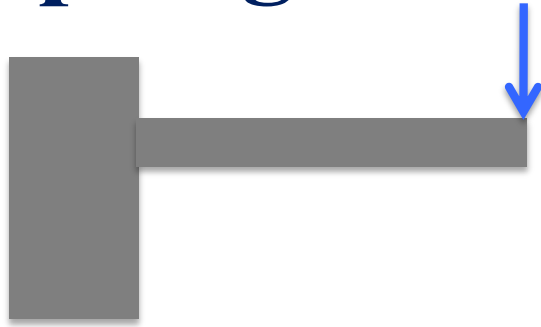
$$\frac{d^2 y}{dx^2} = -\frac{M}{EI}$$

$$= \frac{1}{EI} \cdot F(L - x)$$

B.C.

$$\begin{cases} y(0) = 0 \\ \frac{dy}{dx}(0) = 0 \end{cases}$$

Spring Constant of a Cantilever Beam



$$\frac{d^2 w}{dx^2} = \frac{1}{EI} \cdot F(L - x)$$

$$w = \frac{FL}{2EI} x^2 \left(1 - \frac{x}{3L} \right)$$

$$w_{\max} = w(L) = \frac{L^3}{3EI} \cdot F$$

$$k = \frac{w_{\max}}{F} = \frac{3EI}{L^3} = \frac{WH^3}{4L^3} E$$

When the mass of cantilever is neglected, the resonant frequency becomes

$$\omega = \sqrt{\frac{k}{M}} = \sqrt{\frac{3EI}{ML^3}} = \sqrt{\frac{WH^3}{4ML^3}} E$$

How about ω with a finite mass of the beam?

Derivation with the Energy Method

V : Kinetic energy

T : Strain energy

$$V + T = \text{const.}$$

$$\frac{d}{dt}(V + T) = 0$$

$$\frac{d}{dt}\left(\frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2\right) = 0$$

$$\therefore m\ddot{x}\dot{x} + k\dot{x}x = 0$$

$$\therefore m\ddot{x} + kx = 0$$

$$\therefore \omega = \sqrt{\frac{k}{m}}$$

$V_{\max}$: Kinetic energy

$T_{\max}$: Strain energy

$$V_{\max} = T_{\max}$$

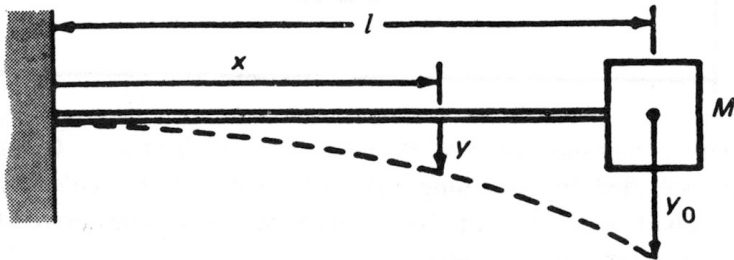
$$\frac{1}{2}m\dot{x}_{\max}^2 = \frac{1}{2}kx_{\max}^2$$

$$x_{\max} = x_0, \dot{x}_{\max} = \omega x_0$$

$$\therefore \frac{1}{2}m(\omega x_0)^2 = \frac{1}{2}kx_0^2$$

$$\therefore \omega = \sqrt{\frac{k}{m}}$$

Derivation with Energy Method



$$k = \frac{y_{\max}}{F} = \frac{3EI}{L^3}$$

$$\therefore 2T = k \cdot y(L)^2 = \frac{3EI}{L^3} y(L)^2$$

Beam shape with applied force F at the tip

$$\begin{aligned} y(x) &= \frac{FL}{2EI} x^2 \left(1 - \frac{x}{3L} \right) \\ &= \frac{y(L)}{2L^3} (3Lx^2 - x^3) \end{aligned}$$

$$2V = \omega^2 \left\{ M \cdot y(L)^2 + \int_0^L \frac{m}{L} y^2 dx \right\}$$

$$\frac{2V}{\omega^2} \approx M \cdot y(L)^2 +$$

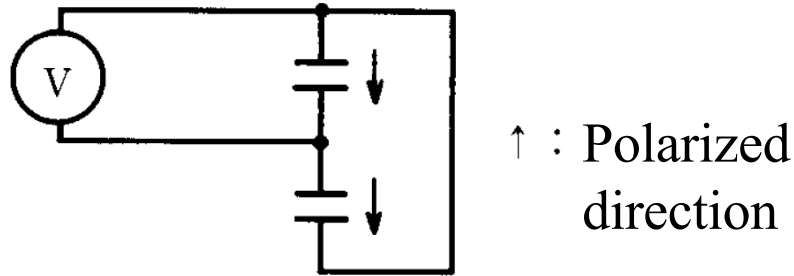
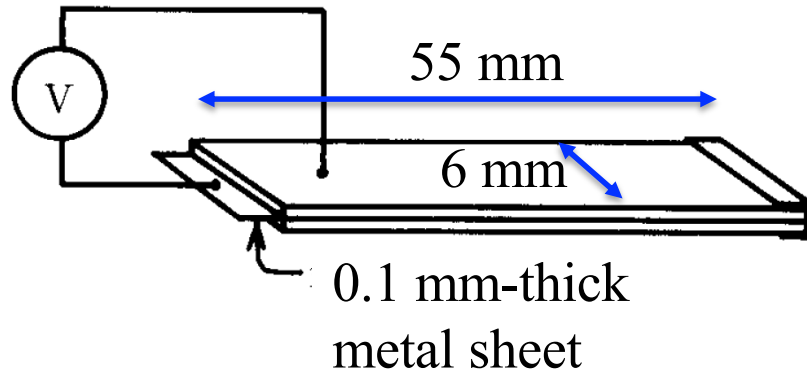
$$\int_0^L \frac{m}{L} \frac{y(L)^2}{4L^6} (3Lx^2 - x^3)^2 dx$$

$$= y(L)^2 \left(M + \frac{33}{140} m \right)$$

With $2T = 2V$

$$\therefore \omega^2 = \frac{3EI}{\left(M + \frac{33}{140} m \right) L^3}$$

Specification of Piezoelectric Cantilever



Total thickness: 0.5 mm

Width: 6 mm

Total length: 55 mm

Material: PZT - $\text{Pb}(\text{Zr} \cdot \text{Ti})\text{O}_3$

Thickness of PZT: 0.2 mm on each side

Young's modulus of PZT: 59 GPa

Density of PZT: 7.75 g/cm^3

Relative permittivity: 5500

Capacitance: 105 nF

$d_{33} = 640 \times 10^{-12} \text{ C/N}$

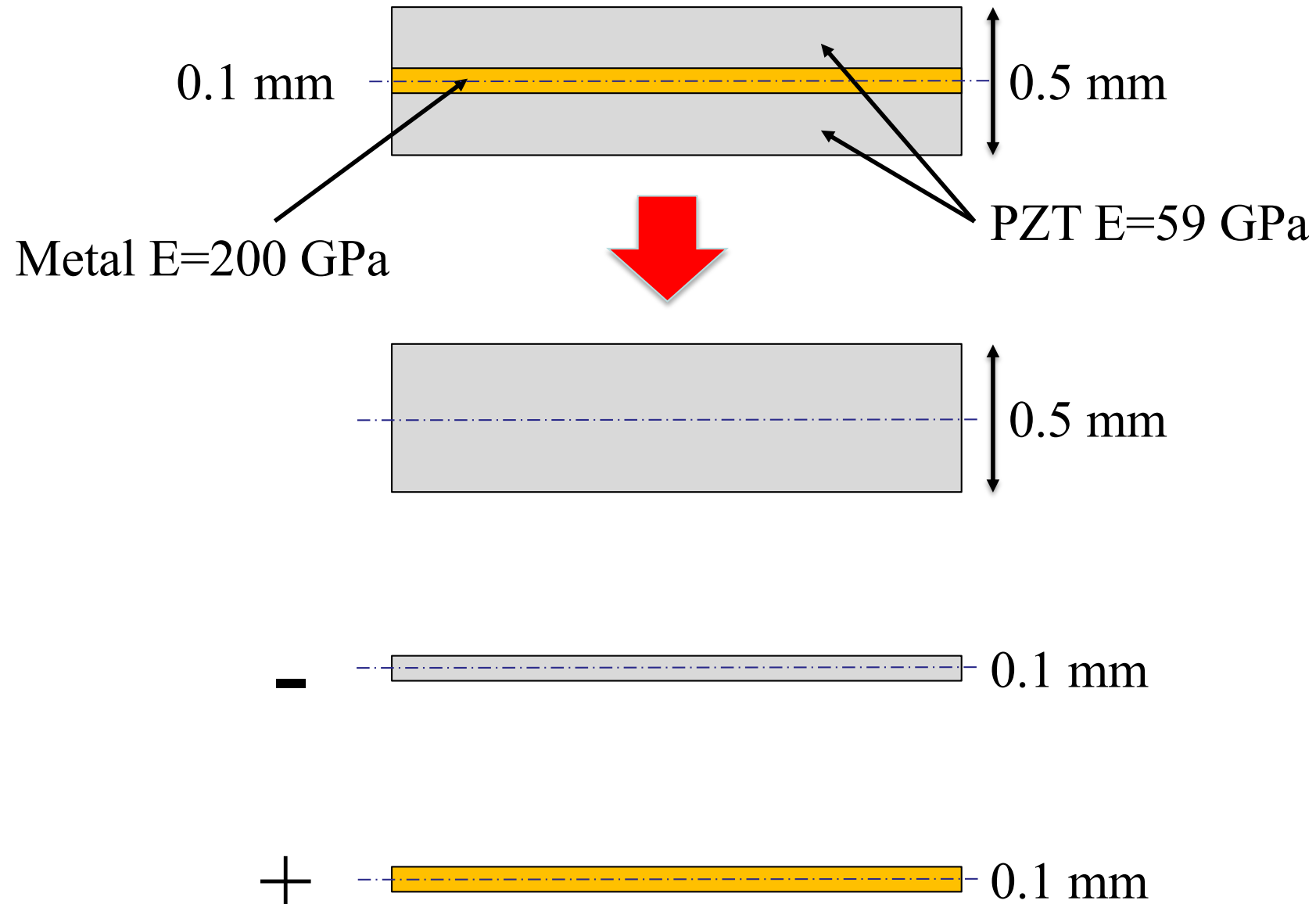
$d_{31} = -330 \times 10^{-12} \text{ C/N}$

Thickness of metal: 0.1 mm

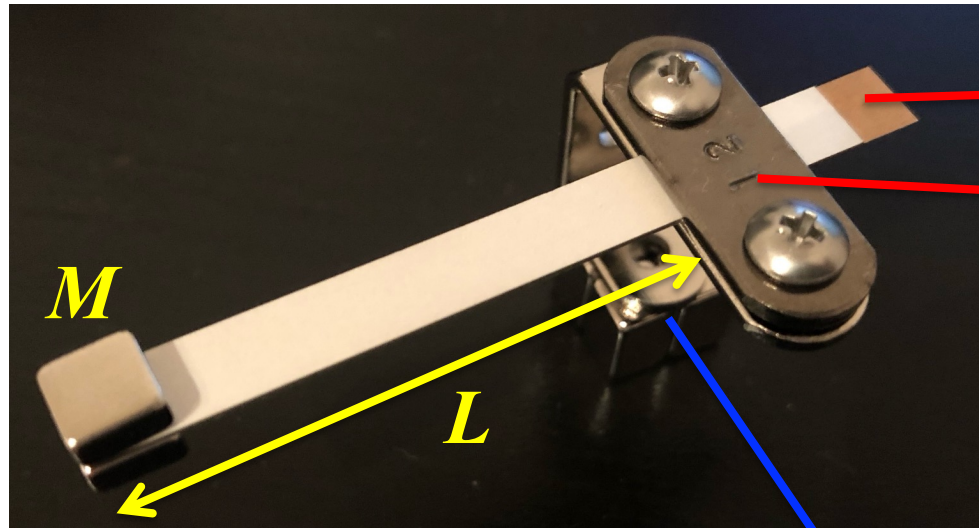
Density of metal: 7.93 g/cm^3

Young's modulus of metal: 200 GPa

Estimate of EI for Multi-layer Structure



Parameters of Piezoelectric Cantilever



Oscillo-
scope

Monitor the voltage
output

L : Length of the
cantilever (including
magnets)

M : Mass of the
magnet (0.81 g x 2)

Accelerometer

Monitor the
acceleration

Amplifier

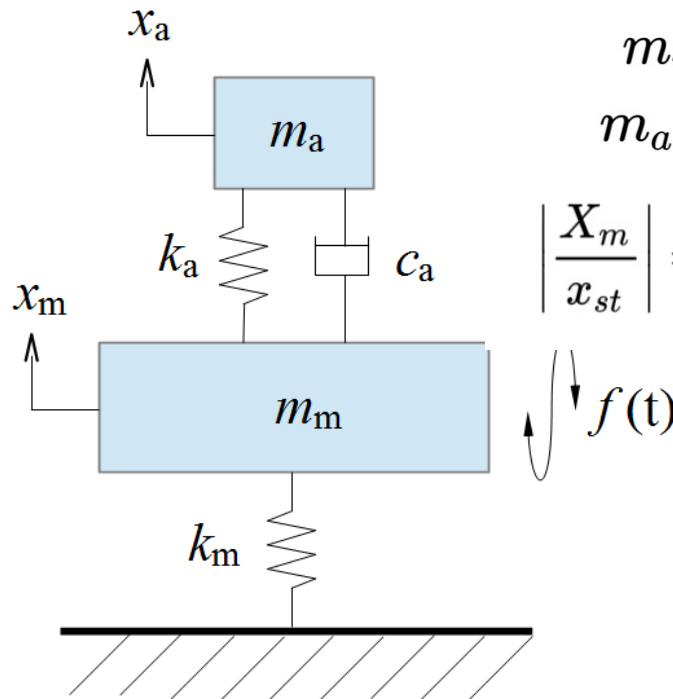
Change frequency and
acceleration



What should be done

- 1) Compute the resonant frequency of the cantilever with a tip mass using the given formula.
- 2) Measurement of the voltage amplitude versus frequency (around 15 – 45 Hz) of the one-mass system. Identify the resonant frequency and compare with the analytical result.
- 3) Measurement of the voltage amplitude versus the external load (around 1 k Ω – 200 k Ω). Compute the output power (V^2/R) at each external load. Identify the optimum load.
- 4) Observe the frequency response of a two-mass system

2 DOF System with Damping

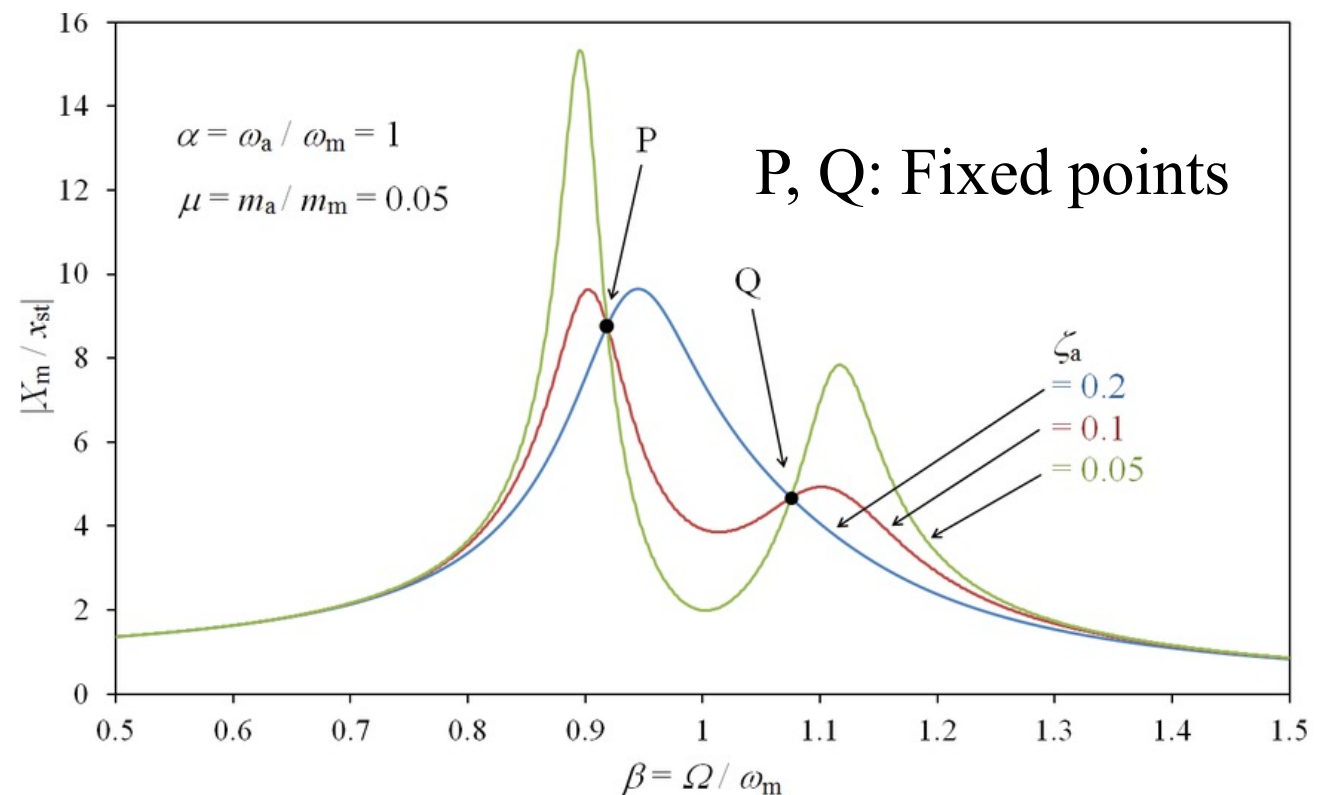


$$m_m \ddot{x}_m + k_m x_m + c_a (\dot{x}_m - \dot{x}_a) + k_a (x_m - x_a) = f_0 e^{i\Omega t}$$

$$m_a \ddot{x}_a + c_a (\dot{x}_a - \dot{x}_m) + k_a (x_a - x_m) = 0$$

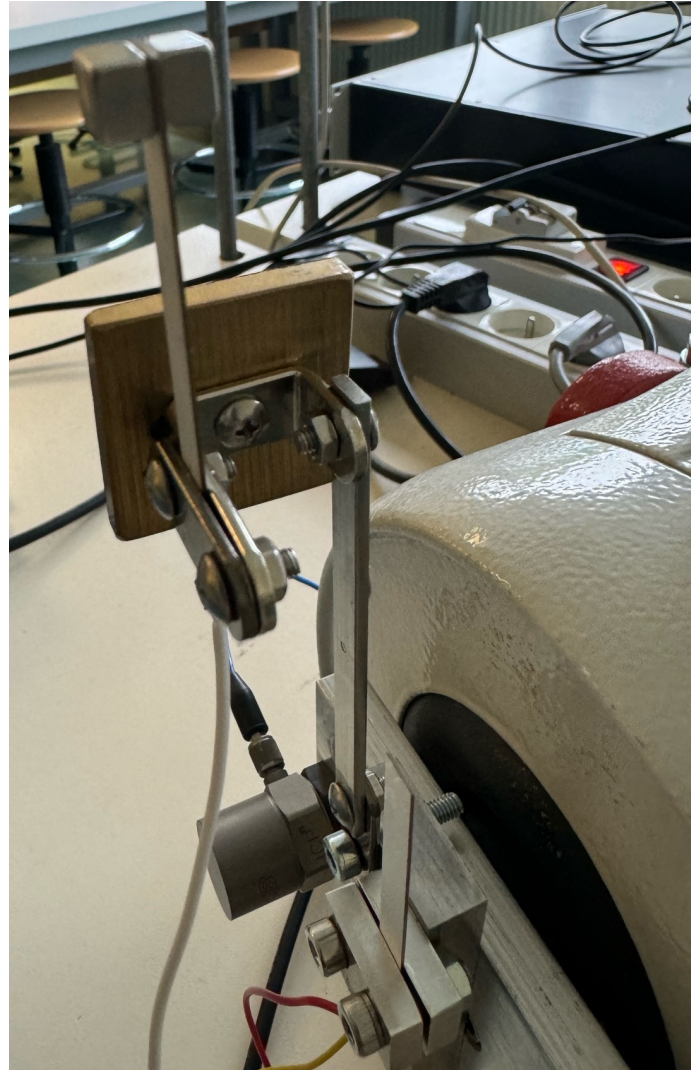
$$\left| \frac{X_m}{x_{st}} \right| = \sqrt{\frac{(\alpha^2 - \beta^2)^2 + (2\zeta_a \alpha \beta)^2}{[(\alpha^2 - \beta^2)(1 - \beta^2) - \mu \alpha^2 \beta^2]^2 + (2\zeta_a \alpha \beta)^2 (1 - \beta^2 - \mu \beta^2)^2}}$$

$$\alpha = \frac{\omega_a}{\omega_m}, \quad \beta = \frac{\Omega}{\omega_m}, \quad c_{ca} = 2\sqrt{m_a k_a}, \quad \zeta_a = \frac{c_a}{c_{ca}}, \quad x_{st} = \frac{f_0}{k_m}$$



Sueoka et al. (2002)
Wikipedia

Demonstration of Broadband EH with Two Mass Approach



Assignment

Produce one PDF/group in your group and send it to me (ysuzuki@mesl.t.u-tokyo.ac.jp) by the end of the day.

- Group number
- Members' name in the group
- Answer the following questions (include the plots)

Q1: Analytical results of the resonant frequency for the cantilever with a tip mass (0.81 g x 2)

Q2: Plot the voltage amplitude versus the oscillation frequency, and identify the resonant frequency. Compare with the results in Q1.

Q3: Plot the output power (V^2/R) versus the external load. Why the output power is maximized at that external load?